

Quick Check

Lesson 3: Observe (Part 2): Lipids and Vitamins — Structure, Function, and Food Tests

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 3

A learner rubs mashed groundnuts on a piece of brown paper bag and waits 5 minutes for it to dry. Which observation shows that the groundnuts contain lipids?

- A** A translucent (greasy) spot stays on the paper even after it is dry
- B** A wet spot appears but disappears completely once the paper dries
- C** The paper stays completely dry and unmarked
- D** The paper turns a purple colour where the food was rubbed

In the ethanol emulsion test, a learner dissolves a food sample in ethanol, then pours the clear ethanol layer into a test tube of distilled water. What result shows that lipid is present?

A The mixture stays clear and colourless

B A cloudy white emulsion forms

C The mixture bubbles and gives off heat

D The mixture turns purple

A group tests ugali paste (maize starch) with both the translucent spot test and the ethanol emulsion test. Which set of results would you expect?

A A permanent translucent spot and a cloudy white emulsion

B No permanent translucent spot but a cloudy white emulsion

C No permanent translucent spot and a clear solution in water

D A permanent translucent spot but a clear solution in water

Which row correctly matches a vitamin with its type and the deficiency disease it causes?

- A** Vitamin C: fat-soluble; deficiency causes scurvy
- B** Vitamin A: fat-soluble; deficiency causes night blindness
- C** B-complex vitamins: fat-soluble; deficiency causes beriberi
- D** Vitamin D: water-soluble; deficiency causes rickets

A child eats a meal rich in vitamin A but has almost no fat in their diet. Why might the child still develop night blindness?

- A** Vitamin A is fat-soluble and needs dietary fat to be absorbed
- B** Vitamin A is a protein, so it cannot be used without amino acids
- C** Fat destroys vitamin A in the stomach
- D** Vitamin A is water-soluble and is washed out of the body too quickly

At the chemical level, what is a lipid (fat) made of?

- A** Vitamins joined to a mineral salt
- B** A glycerol backbone joined to fatty acid chains
- C** Long chains of amino acids joined together
- D** Many glucose units joined in a long chain

Which statement correctly describes the roles of lipids in living organisms?

- A** Lipids are harmful chemicals that the body does not need
- B** Lipids only provide vitamins and have no other role
- C** Lipids store energy and are part of cell membranes
- D** Lipids are the main chemicals used to build muscles and repair tissues

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).